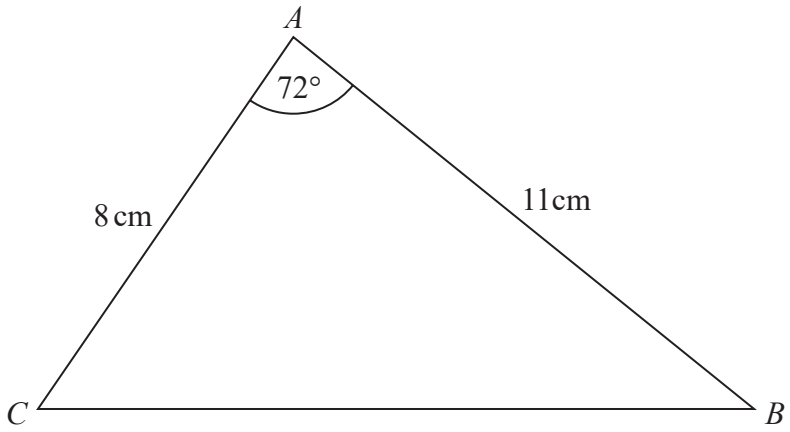
15 Here is triangle *ABC*.



(a) Find the length of *BC*.
Give your answer correct to 3 significant figures.

Cosine rule: $BC^2 = 8^2 + 11^2 - 2(8)(11)\cos 72°$

$BC^2 = 130.613...$

$BC = 11.428... = 11.4$ cm (3 s.f.)

1 mark
correct cosine rule

1 mark
evaluate RHS

1 mark
11.4 cm

11.4 cm

(3)

(b) Find the area of triangle *ABC*.
Give your answer correct to 3 significant figures.

Area $= (1/2)(8)(11)\sin 72°$

$= 41.846... = 41.8$ cm^2 (3 s.f.)

1 mark
half ab sin C

1 mark
41.8 cm2

41.8 cm^2

(2)

(Total for Question 15 is 5 marks)

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